Server

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#define PORT 8080
#define BUFFER_SIZE 1024
int main()
{
    int server_fd, client_socket;
    struct sockaddr_in server_addr, client_addr;
    socklen_t addr_len;
    char filename[100];
    char buffer[BUFFER_SIZE];
    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
    bind(server_fd, (struct sockaddr *)&server_addr,
         sizeof(server_addr));
    listen(server_fd, 5);
    printf("File Server Running...\n");
    while (1)
    {
        addr_len = sizeof(client_addr);
        client_socket = accept(server_fd,
                               (struct sockaddr *)&client_addr,
                               &addr_len);
        if (fork() == 0)
        {
            close(server_fd);
            recv(client_socket, filename, sizeof(filename), 0);
            FILE *fp = fopen(filename, "r");
            sprintf(buffer, "Server PID: %d\n\n", getpid());
            send(client_socket, buffer, strlen(buffer), 0);
            if (fp == NULL)
            {
                strcpy(buffer, "Requested file does not exist.\n");
```

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                send(client_socket, buffer, strlen(buffer), 0);
            }
            else
            {
                while (fgets(buffer, BUFFER_SIZE, fp) != NULL)
                {
                    send(client_socket, buffer, strlen(buffer), 0);
                }

                fclose(fp);
            }
            close(client_socket);
            exit(0);
        }
        close(client_socket);
    }
    return 0;
}
```

Client

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#define PORT 8080
#define BUFFER_SIZE 1024
int main()
{
    int sockfd;
    struct sockaddr_in server_addr;
    char filename[100];
    char buffer[BUFFER_SIZE];
    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    server_addr.sin_addr.s_addr = inet_addr("127.0.0.1");
    connect(sockfd, (struct sockaddr *)&server_addr,
            sizeof(server_addr));
```

```
    printf("Enter filename: ");
    scanf("%99s", filename);
    send(sockfd, filename, strlen(filename) + 1, 0);
    printf("\nResponse from Server:\n\n");
    int len;
    while ((len = recv(sockfd, buffer, BUFFER_SIZE - 1, 0)) > 0)
    {
        buffer[len] = '\0';
        printf("%s", buffer);
    }
    printf("\n");
    close(sockfd);
    return 0;
}
```

Server

```
anandhu@anan:/mnt/e/kichu/Academic/CN lab/exp9$ ./serv
File Server Running...
```

Client

```
anandhu@anan:/mnt/e/kichu/Academic/CN lab/exp9$ ./cli
Enter filename: test.txt

Response from Server:

Server PID: 805

hello this is the file contents.
```